Measuring Waste and Shortage at the Operator Level

A midstream decision layer, and the measurement questions we would value your view on

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WHAT WE ARE BUILDING

We are two Purdue industrial engineers building a supply planning tool for perishable food operators, meaning midstream distributors, aggregators, and food hubs. It reads the order, delivery, and write-off records an operator already keeps, quantifies where margin is lost, and then recommends how much to commit to each supplier and buyer before supply is confirmed. The optimization treats the weekly buying decision as a two stage problem under uncertainty: commit now, absorb the outcome later through spot purchases, waste, or unmet orders.

WHERE THIS SITS RELATIVE TO YOUR WORK

Your research measures waste where it is finally realized, in households. We are working several steps upstream, where we think a meaningful share of it is originally caused. A commitment made two weeks before delivery, against an uncertain harvest, propagates forward: too much and it is written off midstream or pushed into retail and homes on a short remaining shelf life; too little and the buyer is short. That link between an upstream decision and downstream waste is the part of your perspective we most want to test, because if the causal story is weaker than we assume, our framing is wrong.

HOW WE CURRENTLY MEASURE IT, STATED EXACTLY

Waste loss is valued per write-off row at a cost basis, using the row's recorded loss value when present, otherwise quantity lost times the mean unit cost across that product's deliveries. Shortage loss is valued at selling price, as revenue at risk on order lines that were not filled, with substitutions counted as filled. Rows we cannot value are counted and reported rather than silently dropped. The annualized figure is:

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annual_loss = (WasteLoss + ShortageLoss) x 365 / period_days
savings_band = annual_loss x [0.15, 0.30]
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The 15 to 30 percent recovery band is an assumption drawn from USDA SR-77 benchmarks. It is not measured, and we label it as an assumption everywhere it appears.

THE FOUR QUESTIONS WE WOULD MOST VALUE YOUR VIEW ON

- **Mixed valuation bases.** We value waste at cost and shortage at price, then add them into one period loss. Each is defensible alone. Is summing them defensible, or does it need to be reported as two separate numbers?
- **The recovery band.** Is 15 to 30 percent defensible in front of an operator, and is SR-77 the right source, or is there a better grounded estimate in the current literature?
- **Avoidable versus unavoidable.** We do not currently separate the two. How much does that distinction matter when the audience is an operator making a purchasing decision rather than a policy maker?
- **Attribution and boundaries.** If an operator's commitment causes waste that is realized at a retailer or in a home, whose waste is it, and how should a tool like ours account for it honestly?

WHERE WE ARE, PLAINLY

The engine is built and gate tested. Every result so far comes from a calibrated synthetic operator, roughly 8 million dollars and 28 products over 52 weeks. No real operator's records have touched any layer, we have no customers, and we are not raising money. We are recruiting the first operators willing to share six to twelve months of history for a free backtest.

WHAT WE ARE ASKING

Only your critique. You were clear that your connections are household oriented, and we are not asking you to produce introductions. Thirty minutes of pressure testing on the four questions above is worth more to us at this stage, because those numbers will go in front of a real operator and we would rather find out now if we are counting the wrong things.

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